R IS A PSEUDO-TORSOR OVER U X_S U UNDER THE BASE-CHANGED STABILIZER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-groupoid-action-stabilizer (canonical tag 0237). cited at statement lines 3626-3627. Supplies the action of the stabilizer G on R.

Source reference. Original lemma 0237: sources/stacks-chow/tags.txt, source lines 2732–2732. Cited at statement lines 3626-3627: the G-action on R whose restriction induces the G_0-action; load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Let G be the stabilizer group scheme of R. Let*

$$G_0 = G \times_{U, pr_0} (U \times_S U) = G \times_S U$$

as a group scheme over $U \times_S U$. The action of G on R of Lemma 0237 induces an action of G_0 on R over $U \times_S U$ which turns R into a pseudo G_0-torsor over $U \times_S U$.

Proof. This is true because in a groupoid category $\mathcal{C}$ the set $\mathrm{Mor}_{\mathcal{C}}(x, y)$ is a principal homogeneous set under the group $\mathrm{Mor}_{\mathcal{C}}(y, y)$. □

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.